A PROPERTY ETALE LOCAL ON THE SOURCE-AND-TARGET GIVES A PROPERTY OF GERMS THAT IS LOCAL TOO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A germ with property Q admits arbitrarily small representatives with P. Same chapter, Lemma lemma-local-source-target-implies, invoked at proof lines 8086-8090. Allows the proof to replace the existential property by a representative satisfying P on small neighbourhoods.
- The definition of a property etale local on the source-and-target at a point, and the notion of an etale representative of a germ. Same chapter, Definition definition-local-source-target-at-point and Definition definition-etale-morphism-germs, invoked at proof lines 8097-8103. Fixes the shape of the diagram and justifies choosing an etale representative b_1.
- Locality of the largest-open construction for an etale-etale diagram. Same chapter, Lemma lemma-etale-etale-local-source-target, invoked at proof lines 8104-8107. Supplies the equivalence u' in W(h'_1) iff u in W(h_1) that finishes the proof.

Source reference. Original lemma 0CEZ: sources/stacks-descent/pinned-original-source.tex, source lines 7787–7787. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CEY: sources/stacks-descent/pinned-original-source.tex, source lines 6890–6890. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04R0: sources/stacks-descent/pinned-original-source.tex, source lines 7399–7399. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Definition 0.1. Let $f : (X, x) \to (S, s)$ be a morphism of germs. We say f is *étale* (resp. *smooth*) if there exists a representative $f : U \to S$ of f which is an étale morphism (resp. a smooth morphism) of schemes.

Definition 0.2. Let $\mathcal{Q}$ be a property of morphisms of germs of schemes. We say $\mathcal{Q}$ is *étale local on the source-and-target* if for any commutative diagram

$$\begin{array}{ccc} (U', u') & \xrightarrow[h']{} & (V', v') \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ (U, u) & \xrightarrow{h} & (V, v) \end{array}$$

of germs with étale vertical arrows we have $\mathcal{Q}(h) \Leftrightarrow \mathcal{Q}(h')$.

Lemma 0.3. *Let $\mathcal{P}$ be a property of morphisms of schemes which is étale local on the source-and-target. Consider the property $\mathcal{Q}$ of morphisms of germs defined by the rule*

$$\mathcal{Q}((X, x) \to (S, s)) \Leftrightarrow \textit{there exists a representative } U \to S \textit{ which has } \mathcal{P}$$

Then $\mathcal{Q}$ is étale local on the source-and-target as in Definition 0.2.

Proof. If a morphism of germs $(X, x) \to (S, s)$ has $\mathcal{Q}$, then there are arbitrarily small neighbourhoods $U \subset X$ of x and $V \subset S$ of s such that a representative $U \to V$ of $(X, x) \to (S, s)$ has $\mathcal{P}$. This follows from Lemma 04R0. Let

$$\begin{array}{ccc} (U', u') & \xrightarrow[h']{} & (V', v') \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ (U, u) & \xrightarrow{h} & (V, v) \end{array}$$

be as in Definition 0.2. Choose $U_1 \subset U$ and a representative $h_1 : U_1 \to V$ of h. Choose $V_1' \subset V'$ and an étale representative $b_1 : V_1' \to V$ of b (Definition 0.1). Choose $U_1' \subset U'$ and representatives $a_1 : U_1' \to U_1$ and $h_1' : U_1' \to V_1'$ of a and h' with a_1 étale. After shrinking U_1' we may assume $h_1 \circ a_1 = b_1 \circ h_1'$. By the initial remark of the proof, we are trying to show $u' \in W(h_1') \Leftrightarrow u \in W(h_1)$ where $W(-)$ is as in Lemma 0CEY. Thus the lemma follows from Lemma 0CEZ. □